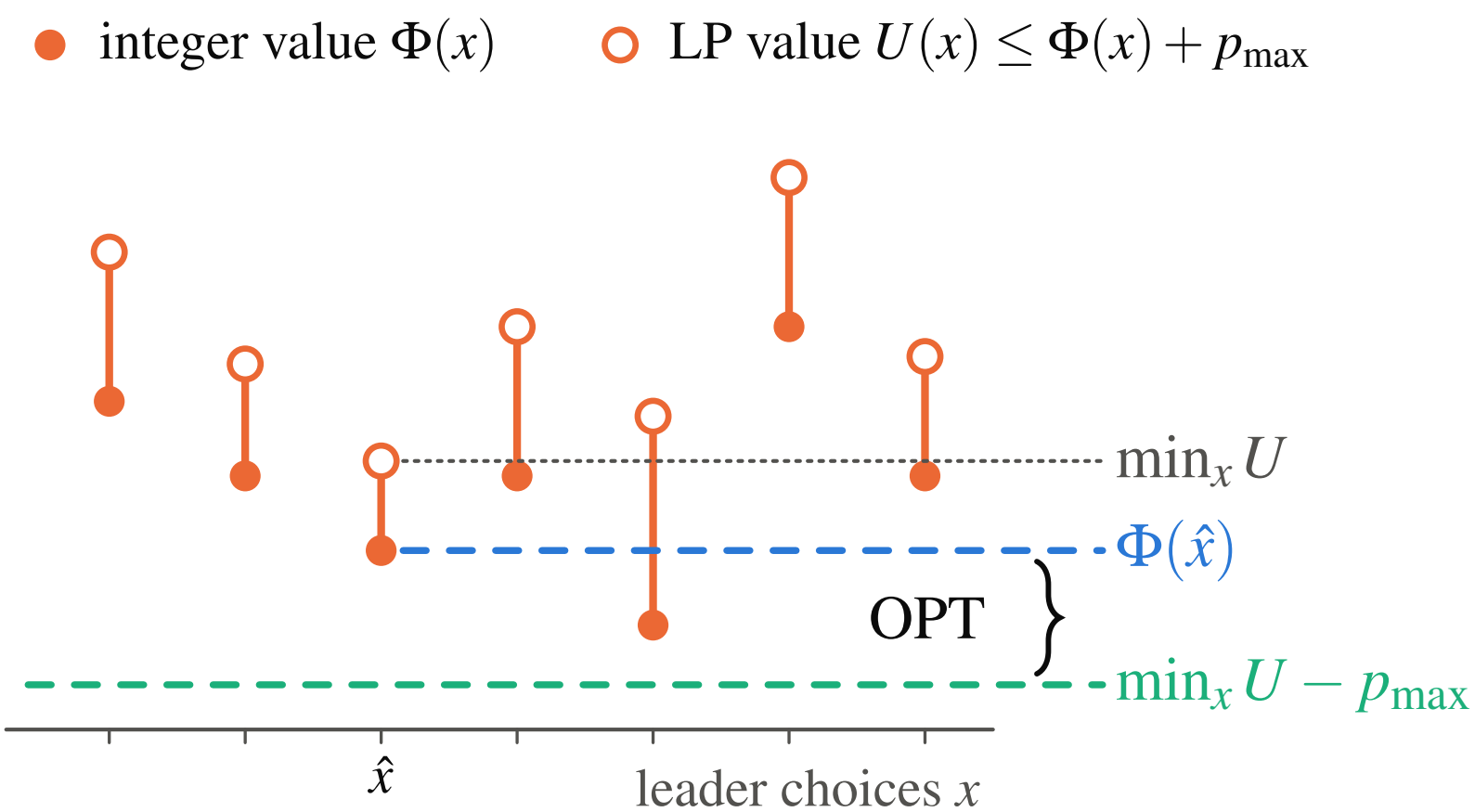
integer value $\Phi(x)$
LP value $U(x) \leq \Phi(x) + p_{\max}$
$\min_x U$
$\Phi(\hat{x})$
OPT
$\min_x U - p_{\max}$
$\hat{x}$
leader choices x